

Task-Time Morphology–Motion Co-Design under Reconfiguration Feasibility

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Abstract—Self-reconfigurable robots can change embodiment during a task, which creates a design decision that fixed-morphology robots never face: *where can the robot physically transform?* We formulate task-time co-design over route, morphology sequence, and reconfiguration site, and ask whether modeling the physical feasibility of the transformation itself changes which design is selected. On a simulated six-pod self-mobile robot with two qualified morphologies, we compare route-first adaptation, geometry-coupled search, and feasibility-coupled search in diagnostic environments. In nine end-to-end missions, geometry-coupled search chose sites where both morphologies fit but a relocating pod collided with an obstruction, and both of its transformations there failed safely; every transformation at a site accepted by the feasibility model executed without contact, and all planners agreed in the unobstructed control. Preliminary as they are, the results indicate that where a robot can physically change embodiment should be part of the co-design decision.

I. MOTIVATION AND POSITIONING

Robot co-design has largely optimized a body *before* deployment, from joint evolution of morphology and control [1] to grammar-based and learned design search [2], [3]. Self-reconfigurable modular robots instead choose embodiment *during* a task. Prior work already selects configurations to satisfy task and environment requirements [4], [5], plans paths across a discrete set of morphologies [6], and plans module motions or connectivity changes from one configuration to another [7], [8]. Joint morphology–navigation planning is therefore not our claim. Task-level planners commonly treat a transformation as a switch between two admissible configurations, while reconfiguration planners take the target and its location as given. The coupling between the two—two configurations can each be collision-free at a location while the module motions that connect them are not—falls between these layers.

We study this coupling as *task-time iterative co-design*: embodiment is an optimization variable coupled to how and where the robot moves, over route $\times$ morphology $\times$ reconfiguration site $\times$ transition feasibility. Learned morphology search [2], [3] is complementary: it expands the design space, whereas we ask whether and where an available embodiment change can physically occur within a mission.

II. PLATFORM AND PLANNERS

The robot is a sensing core with six independently mobile differential-drive pods in Gazebo/ROS 2 (Fig. 1a). Two morphologies are qualified end to end: `compact_diff` and a longer, narrower `narrow_tandem` for constrained passages. Reconfiguration is executed, not teleported: pods detach, drive

along declared relocation paths, align using connector observations, and latch; execution tracks observed topology and inhibits assembled motion until a valid commit. The planner searches over (x, y, θ, m) . Traversal edges use morphology-specific footprints and costs; transition edges carry a pre-action failure probability.

Three planners share the same mechanics, controllers, map, and disturbance realization: **Route-first** fixes a spatial route and then inserts morphology changes along it. **Geometry-coupled** searches route and morphology jointly, admitting a transition when a planar clearance disk and the target footprint are free. **Feasibility-coupled** additionally samples every relocating module’s 3D collision boxes along its sequential trajectory against the environment, stationary core, and support and latch conditions, and rejects the edge with a structured reason. Thus route-first $\rightarrow$ geometry-coupled isolates coupling route and embodiment, and geometry-coupled $\rightarrow$ feasibility-coupled isolates physical transition feasibility.

III. DIAGNOSTIC STUDY

One passage environment is generated by shared code in three variants. The doorway admits only `narrow_tandem`. *Blocked-A* places a low post beside the attractive staging site so that before/after footprints remain admissible but the outward relocation of one pod intersects it; *Blocked-B* mirrors the post to intercept a different pod; *Neutral* removes it. The post lies below the lidar plane and outside the assembled robot’s corridor, so it constrains only the transformation. Before simulation, deterministic planner-only checks confirmed the intended manipulation (Fig. 1b,c): in both blocked variants geometry-coupled search staged at $x = 2.15$ m, while feasibility-coupled search rejected 711 (A) and 1697 (B) edge evaluations for module–environment collision and staged at $x = 1.85$ m; in Neutral all three planners staged at $x = 2.15$ m with no environment rejections. We then ran nine full missions (3 variants $\times$ 3 planners) under one shared disturbance realization (ground friction and initial pose offset) through the complete Gazebo, Nav2, localization, reconfiguration, and evaluator stack, with immutable per-trial records and provenance. Evaluator-only Gazebo poses supply clearance and contact; no planner or controller consumes them.

IV. PRELIMINARY RESULTS

The feasibility term changed the design and predicted the physical outcome. In Blocked-A, geometry-coupled planning staged where both assembled morphologies fit (Fig. 1b)

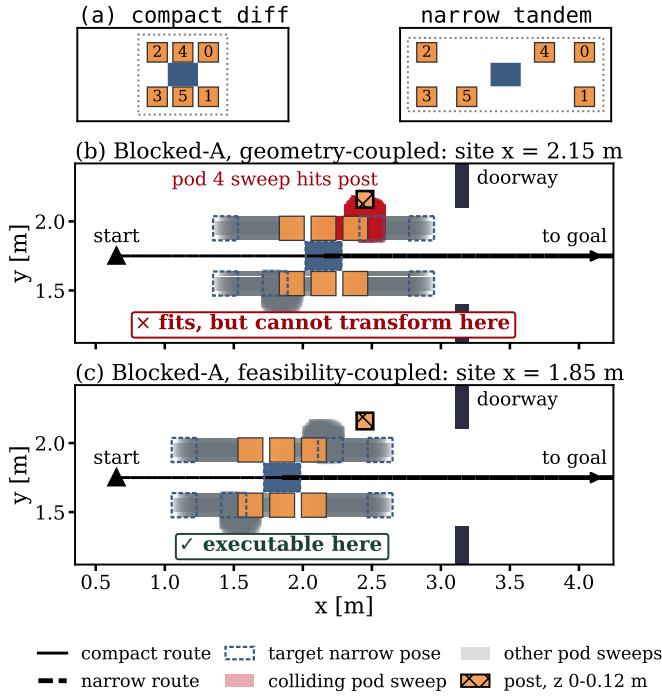


Fig. 1. (a) The two qualified morphologies (core in blue, pods numbered; dashed: planning footprint). (b) Blocked-A, drawn from planner and scenario geometry only. A low post (orange) lies outside the assembled robot’s corridor but inside pod 4’s outward *compact_to_narrow* relocation sweep (red) at the site preferred by geometry-coupled search ($\times$). Feasibility-coupled search rejects that edge and stages earlier ($\circ$); route-first adaptation, which checks the same trajectory feasibility along its fixed route, does likewise ($\square$).

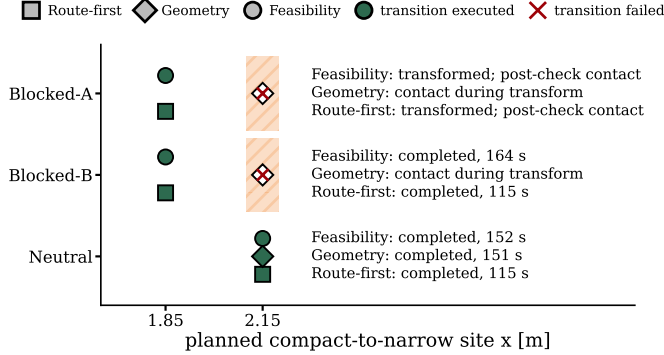


Fig. 2. Preliminary end-to-end evidence: one mission per environment and planner ($n = 1$ per marker; not rates). Marker position is the executed *compact_to_narrow* site; fill indicates the executor committed the transformation, $\times$ that it failed. Hatched band: site intercepted by the post. Text gives mission outcome; contact phases come from evaluator-only Gazebo poses.

and assigned the transition a 0.955 predicted success. During execution pod 4 remained in contact with the post for 40 s of relocation, the transformation did not complete, and the robot entered *RECOVERY_REQUIRED* with assembled drive inhibited. Feasibility-coupled planning rejected that edge and transformed 0.3 m earlier; the executor committed *narrow_tandem* with no contact during the transformation. Route-first adaptation, which applies the same trajectory check

along its fixed route, chose the same site and transformation outcome. Blocked-B repeated the pattern with the other pod: geometry-coupled staging again failed (pod 5 in contact for 42 s, *RECOVERY_REQUIRED*), whereas feasibility-coupled planning transformed at $x = 1.85$ m and completed the mission in 164 s with 4.7 cm minimum clearance and no contact. Route-first adaptation also completed (115 s). In Neutral all three planners staged at $x = 2.15$ m, logged no environment rejections, and completed with 3.7–4.5 cm clearance, so the feasibility term created no artificial difference (Fig. 2). Across the nine missions, all seven *compact_to_narrow* transformations at sites the trajectory check accepts executed without contact, while both geometry-accepted transformations at obstructed sites failed. The added reasoning is not free: the coupled planners took 23–52 s per mission versus about 1 s for route-first adaptation, which searches fewer joint states.

An unmodeled post-transition maneuver. Neither transformed mission in Blocked-A completed: 0.4 s into the automatic post-transition motion check, an in-place yaw of the 1.6 m-long *narrow_tandem* body swung pod 0 into the post, and the check inhibited drive. The transition model covers module relocation but not this verification motion, so the evidence supports feasibility-aware site selection for the transformation itself while exposing that every motion performed at the site belongs in the feasibility model.

V. DISCUSSION AND LIMITATIONS

Preliminary evidence suggests that morphology selection and morphology transition cannot always be separated: a site can accommodate both the pre- and post-transition robot and still be unable to host the transformation, so the site and physical executability of embodiment change should be first-class variables in task-time co-design. The same logic extends beyond relocation to every motion performed at the site. These are diagnostic cases, not a statistical comparison: two discrete morphologies, simulation only, targeted environments, one disturbance realization per case, idealized connectors and latches, and static 3D obstacles supplied to the planner as a known prior map. Next steps are uncertain sensing of transition sites, learned transition-outcome models trained from executed transformations, larger morphology spaces, and hardware. Coupling generative morphology proposals with explicit transition feasibility may let robots decide not only *what* to become, but *when and where* they can become it.

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